

Name: _____

Class: _____



Mr. Rawson • GCSE Computer Science

Year 11 2026

Programming Challenge: WHILE Loops

Sum of Positive Numbers (HARD) — Answer Key

https://isaacomputerscience.org/questions/whileloops_hard

- Write a Python program that repeatedly asks the user for numbers and adds them together.
- The program stops asking when the user enters a negative number and then prints the total sum of the positive numbers.

Example Output:

```
> Enter a number: 5
> Enter a number: 12
> Enter a number: -1

The sum of positive numbers is: 17
```

Previous Knowledge Needed

- A running total (line 2)
- WHILE loops (line 10)
- Input converted with int() (line 6)

Alfred's answer, v2.

```
1 # the total sum = a running total, starting at 0
2 total = 0
3
4 # repeatedly asks the user for numbers = int(input())
5 # ask once BEFORE the loop, so the loop has a number to check
6 number = int(input("Enter a number: "))
7
8 # stops asking when the user enters a negative number
9 # = keep going while the number is 0 or more
10 while number >= 0:
11
12     # adds them together = total = total + number
13     # for 5, 12: 0+5=5, then 5+12=17
14     total = total + number
15
16     # asks again = int(input()), at the END of the loop
17     # a negative number here makes the while check False, so the loop stops
18     number = int(input("Enter a number: "))
19
20 # then prints the total sum of the positive numbers = print(), AFTER the loop
21 # The sum of positive numbers is: 17 = print(..., total)
22 print("The sum of positive numbers is:", total)
```